

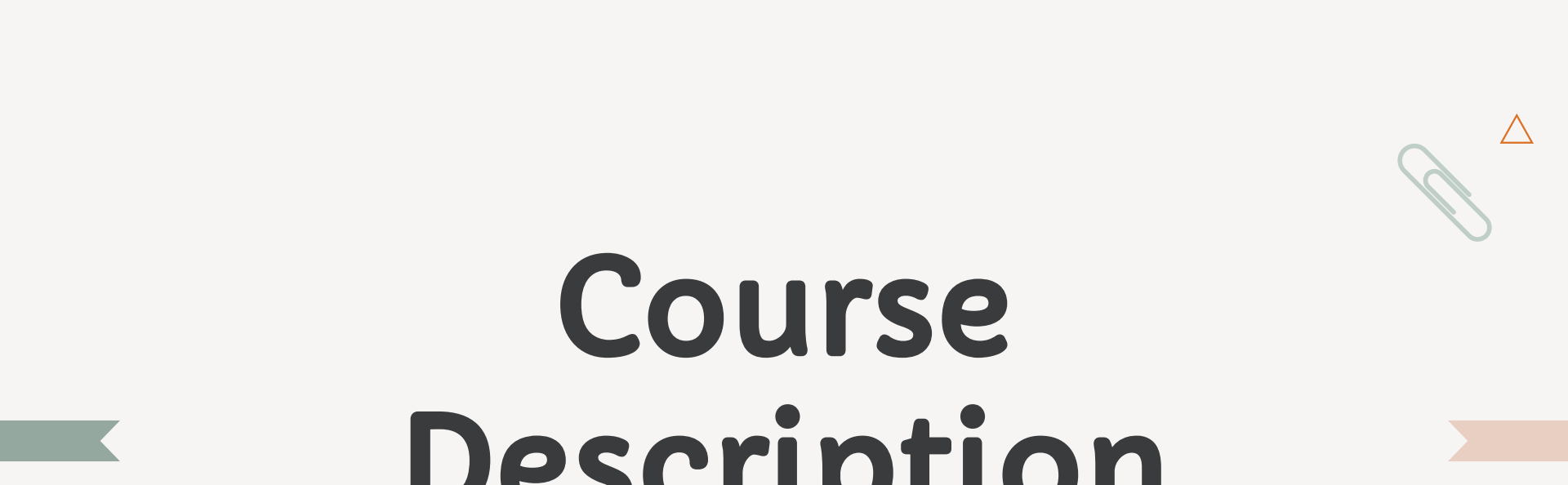


Introduction

Department of Computer Engineering
Sharif University of Technology

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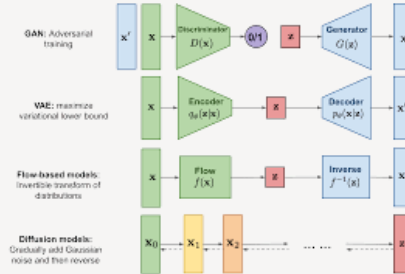
Maryam Ramezani maryam.ramezani@sharif.edu



Course Description

How is our course?







Machine Learning Mathematics

01

Linear Algebra

Provides the foundation for manipulating data in high-dimensional spaces, essential for vector operations in machine learning.

02

Statistics & Probability

Offers tools to model uncertainty and make inferences about data, forming the backbone of many machine learning algorithms.

03

Optimization

Focuses on finding the best parameters for a model by minimizing or maximizing an objective function.

04

Information Theory

Analyzes data compression and transmission, helping quantify uncertainty and information gain in machine learning models.

05

Signal Processing

Involves analyzing and transforming data signals, important for handling time-series data and feature extraction in machine learning.

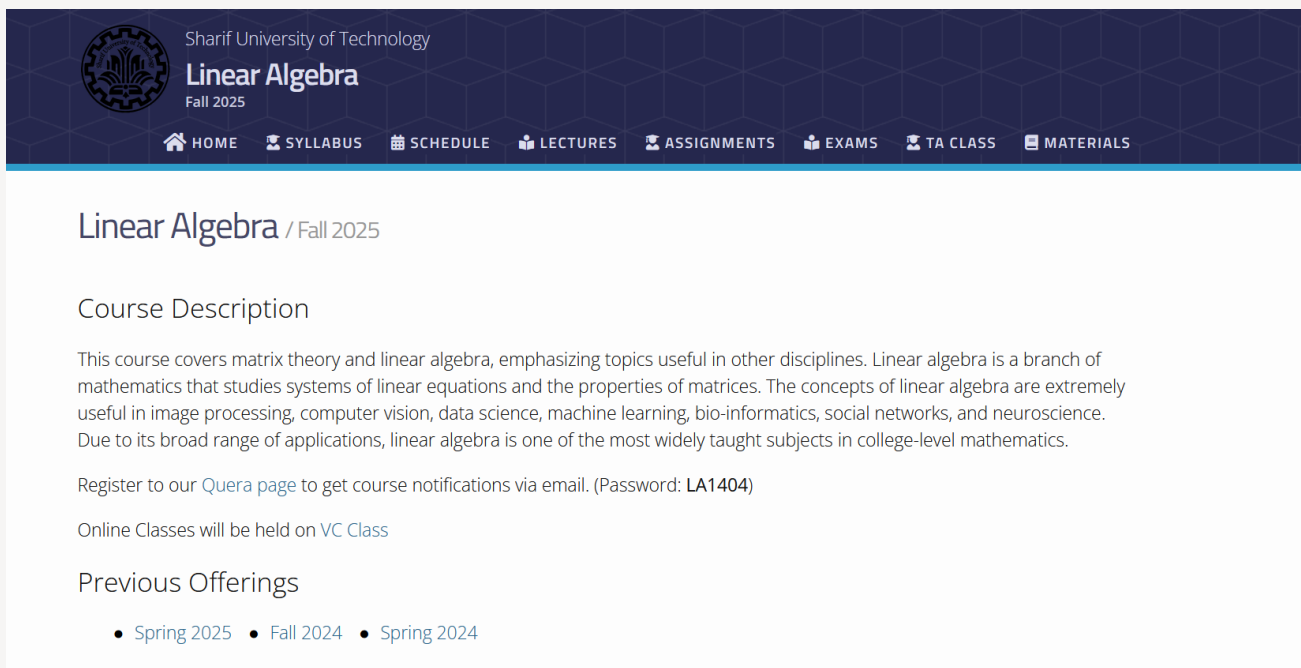
06

Graph Theory

Studies networks of nodes and edges, crucial for understanding relationships in data and complex models like Graph Neural Networks.

Course Material

<https://sut-ce-courses.github.io/linearalgebra> (jabrekh.ir)



The screenshot shows the homepage of the 'Linear Algebra' course website for Fall 2025. The header features the Sharif University of Technology logo and the course title 'Linear Algebra Fall 2025'. A navigation bar includes links for HOME, SYLLABUS, SCHEDULE, LECTURES, ASSIGNMENTS, EXAMS, TA CLASS, and MATERIALS. The main content area has a heading 'Linear Algebra / Fall 2025' followed by a 'Course Description' section. The description states that the course covers matrix theory and linear algebra, emphasizing applications in image processing, computer vision, data science, machine learning, bio-informatics, social networks, and neuroscience. It also mentions that linear algebra is one of the most widely taught subjects in college-level mathematics. Below the description, there is a registration link for a 'Quera page' with the password 'LA1404' and a note that online classes will be held on 'VC Class'. The 'Previous Offerings' section lists 'Spring 2025', 'Fall 2024', and 'Spring 2024'.

Sharif University of Technology

Linear Algebra

Fall 2025

[HOME](#) [SYLLABUS](#) [SCHEDULE](#) [LECTURES](#) [ASSIGNMENTS](#) [EXAMS](#) [TA CLASS](#) [MATERIALS](#)

Linear Algebra / Fall 2025

Course Description

This course covers matrix theory and linear algebra, emphasizing topics useful in other disciplines. Linear algebra is a branch of mathematics that studies systems of linear equations and the properties of matrices. The concepts of linear algebra are extremely useful in image processing, computer vision, data science, machine learning, bio-informatics, social networks, and neuroscience. Due to its broad range of applications, linear algebra is one of the most widely taught subjects in college-level mathematics.

Register to our [Quera page](#) to get course notifications via email. (Password: **LA1404**)

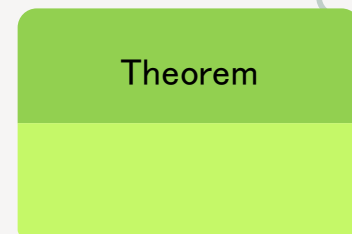
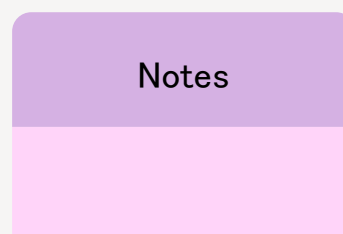
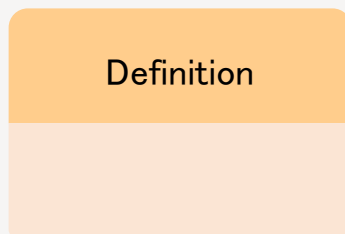
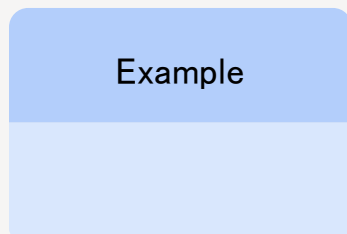
Online Classes will be held on [VC Class](#)

Previous Offerings

- [Spring 2025](#)
- [Fall 2024](#)
- [Spring 2024](#)

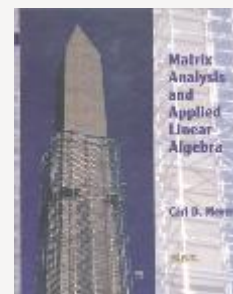
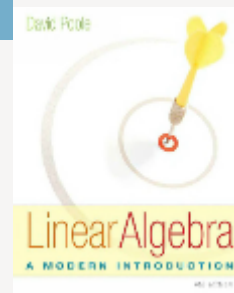
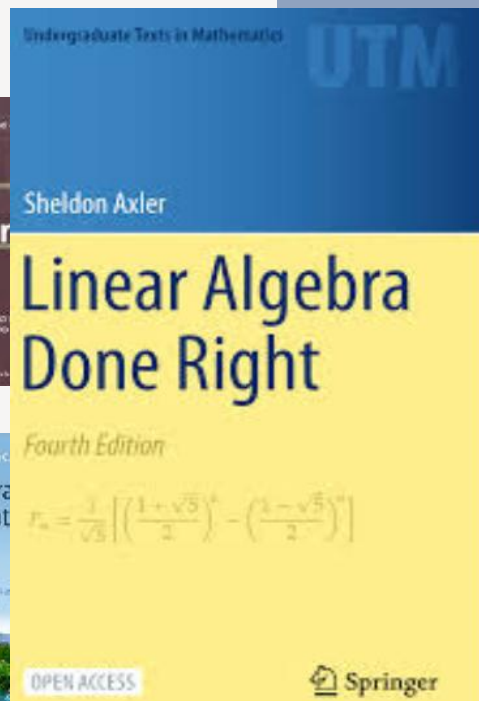
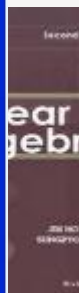
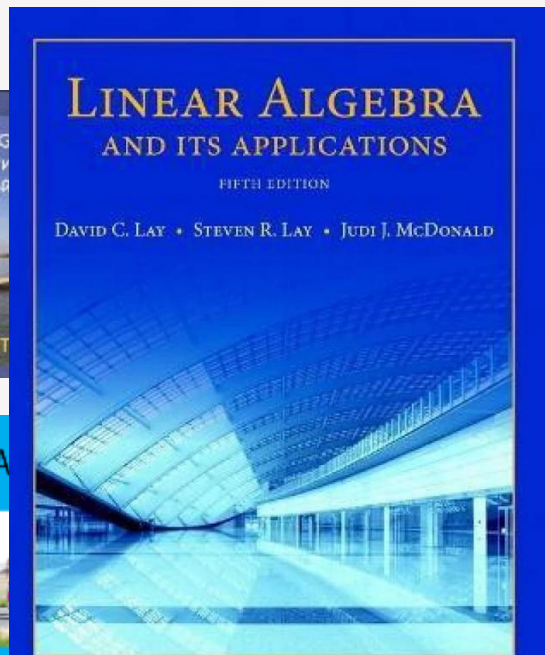
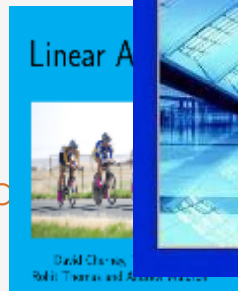
Course Method

- Slides



- Writing on board
- Geometric Interpretation and Intuition
- Notebooks

Course References



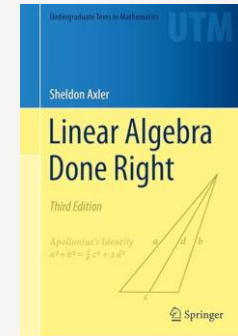
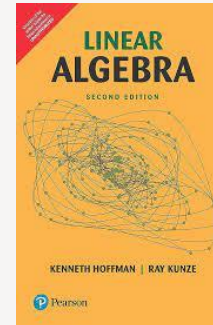
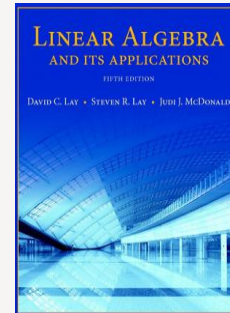
Reach Us!

- Instructors:
 - Hamid R. Rabiee: (CE804) rabiee@sharif.edu
 - Maryam Ramezani: (CE821) maryam.ramezani@sharif.edu
- Course notes, homework and solutions, handouts, and other useful resources are available on the course page:
 - [https://quera.org/course/add_to_course/course/27137/\(password: La1125Spring\)](https://quera.org/course/add_to_course/course/27137/(password:La1125Spring))
 - Room: (Sunday & Tuesday: 10:30-12:00)
 - Ebnesina Alef 11
 - <https://vc.sharif.edu/ch/maryam.ramezani> & <https://vc.sharif.edu/ch/rabiee>
- Lead TAs:
 - Amirmahdi Vaziri (Amirmahdi.vaziri2005@gmail.com, ID : [@kiavzr](#))

Resources

- Textbooks:
 - Sheldon Axler, Linear Algebra Done Right, fourth edition, 2023
 - Kenneth Hoffman and Ray A. Kunze. Linear Algebra. PHI Learning, 2004.
 - Gilbert Strang. Introduction to Linear Algebra. Wellesley-Cambridge Press, 2016.
 - David C. Lay, Steven R. Lay, and Judi J. McDonald. Linear Algebra and Its Applications. Pearson, 2016.

+Other textbooks and course materials.



Structure of the Course

Lectures

Goal: To introduce concepts in linear algebra, and motivate their use and importance.

Note: We try to cover useful materials in class, but we recommend you reading more!

Exams

Quiz: 5 with dropping lowest

Midterm: 1

Final: 1

Assignments

Purpose: To give you a chance to exercise your mind, and to solidify the concepts introduced to you in class.

Structure: Seven theoretical assignments, and Two linear algebra practical assignments.

Importance: Not important unless you want to learn the material and get a good grade.

Number of Exercises: Written: 7
Programming: 2

Grading Policy

Quizzes

3.6 points (5 quizzes each 0.9 points; the lowest score will be dropped)

Theoretical Assignments

3.5 points (each 0.5 points)

Programming Assignments

2 points (each 1 point)

Midterm

6 points (Lectures part 1,2,3,4)

Final

6 points (Lectures part 5,6,7)

Total: 21.1 Points

Lecture Notes

- Lecture slide will be uploaded.
- Many times we will write on board, in real-time, during lecture to prove a theory or answer a question or add some additional explanations. It will be your responsibility to take notes.
- Slides links will be provided on site.

Syllabus

1404/12/03	S01	1	Introduction
1404/12/05	S02		Elementary Row Operations, and Linear Equations
1404/12/10	S03		Row Reduction, Echelon Forms and LU Factorization
1404/12/12	S04		Vector Space
			Subspace

1404/12/17	S05	2	Linear Independence
1404/12/19	S06		
1404/12/24	S07		Bases, Dimension
1405/01/16	S08		
1405/01/18	S09		Matrix Rank

Syllabus

1405/01/23	S10	3	Linear Transformation
1405/01/30	S11		
1405/02/01	S12		Change Basis
1405/02/06	S13		Inverse

1405/02/08	S14	4	Inner Product Space and Norm2
1405/02/13	S15		
1405/02/15	S16		Orthogonality (Gram–Schmidt, QR , etc.)

Syllabus

1405/02/20	S17	5	Determinant
1405/02/22	S18		
1405/02/27	S19		Eigenvectors and Eigenvalues
1405/02/29	S20		Similarity & Diagonalization
1404/02/31	Midterm Exam		

1405/03/03	S21	6	Symmetric Matrices and Quadratic Forms
1405/03/05	S22		Matrix Factorization
1405/03/10	S23		Singular Values and Singular Vectors
1405/03/12	S24		SVD

Syllabus

1405/03/17	S25	7	Norm Space
1405/03/19	S26		Vector and Matrix Derivatives
1405/03/24	S27		
1405/03/26	S28		Least squares
1405/04/16	Final Exam		

Assignments

- Theoretical assignments are released at the end of each part. 3 questions of each assignment are homework. Each has 0.5 points
- Programming Assignments: 2 points

Assignments

Assignment	Release	Deadline	Solution Release
T1	1404/12/12	1404/12/17	1404/12/18
T2	1405/01/18	1405/01/25	1405/01/26
T3	1405/02/06	1405/02/13	1405/02/14
T4	1405/02/15	1405/02/22	1405/02/23
T5	1405/02/29	1405/03/05	1405/03/06
T6	1405/03/12	1405/03/19	1405/03/20
T7	1405/03/26	1405/04/05	1405/04/06

Exams

Quizzes will be held as the following table on 12:00 for half an hour and will be graded in Grade Scope.

Exam	Date
Quiz 1	1404/12/19
Quiz 2	1405/02/01
Quiz 3	1405/02/20
Midterm	1405/02/31
Quiz 4	1405/03/10
Quiz 5	1405/3/24
Final	1405/04/16

TA Classes

TA classes will be held live.





Introduction

Lets think about a question!

01

Vector

Basic Concept

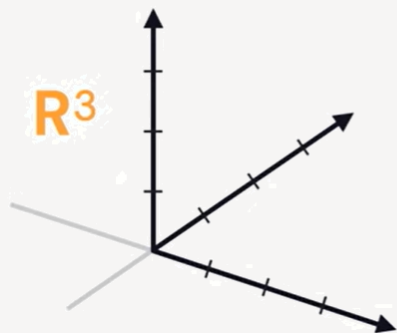


Data Representations (Linear Algebra)

- How can we represent data (images, text, user preferences, etc.) in a way that computers can understand?
- Organize information into a vector!
 - A vector is a 1-dimensional array of numbers.
 - It has both a magnitude (length) and a direction
- The totality of a vectors with n entries is an n-dimensional vector space


$$\vec{V} = \begin{bmatrix} -3 \\ 0.7 \\ 2 \end{bmatrix}$$

“3-dimensional space” consists
of all vectors with 3 entries:



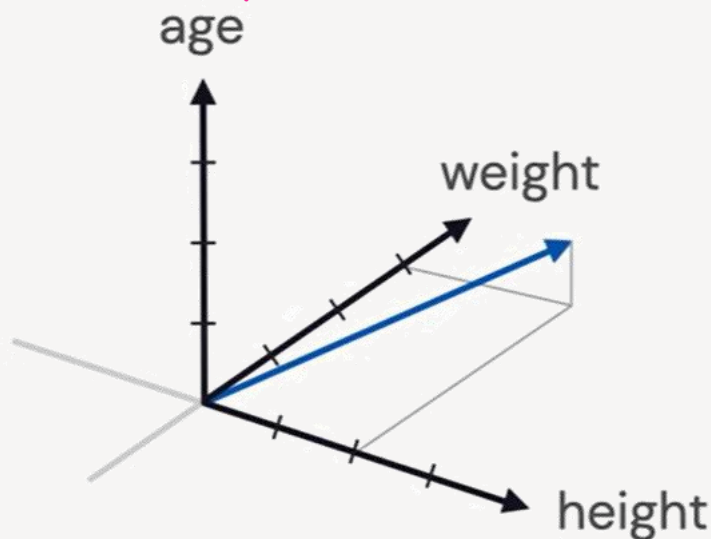
$$\begin{bmatrix} * \\ * \\ * \end{bmatrix}$$



Data Representations (Machine Learning)

- A **feature vector** is a vector whose entries represent the “features” of an object.
- The vector space containing them is called **feature space**.

$$P = \begin{bmatrix} 64 \\ 131 \\ 24 \end{bmatrix} \begin{matrix} \text{height} \\ \text{weight} \\ \text{age} \end{matrix}$$





02

Equation with Matrix and Vector Format



House Pricing Example



Price Problem



\$ 70'000



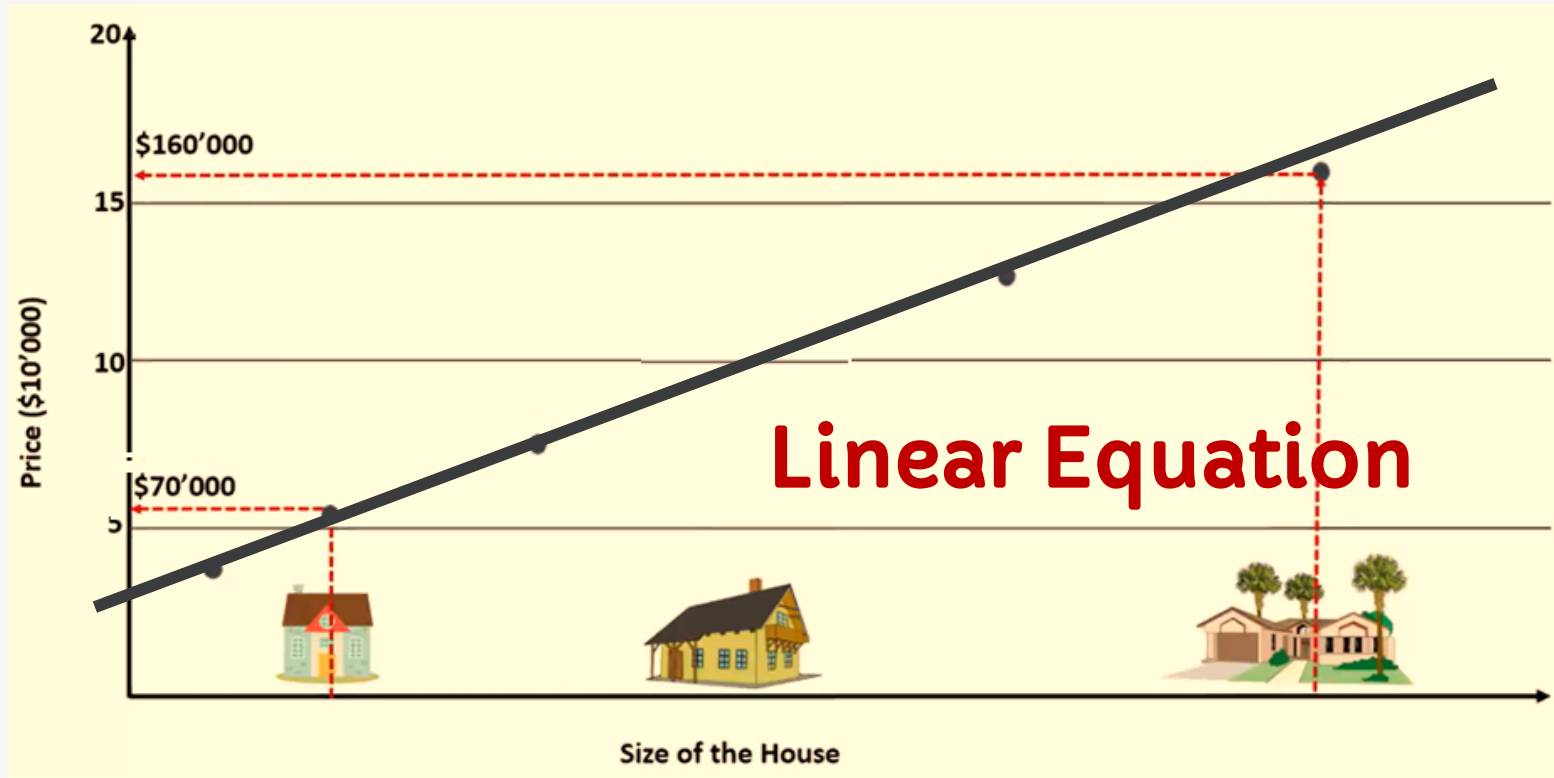
?



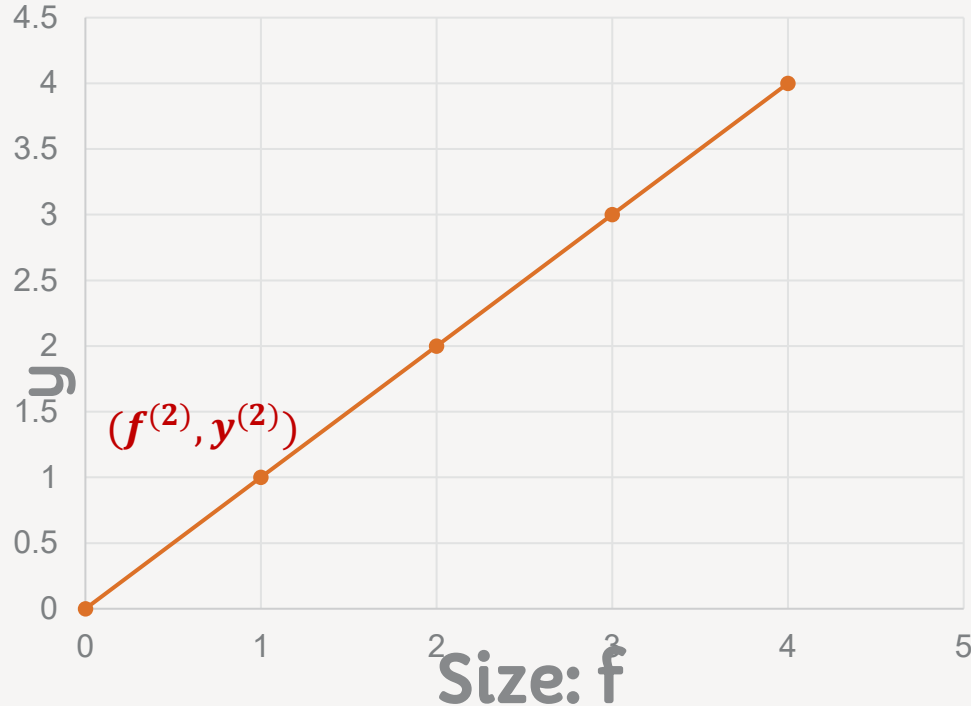
\$ 160'000



Price Problem



Linear Equation without offset

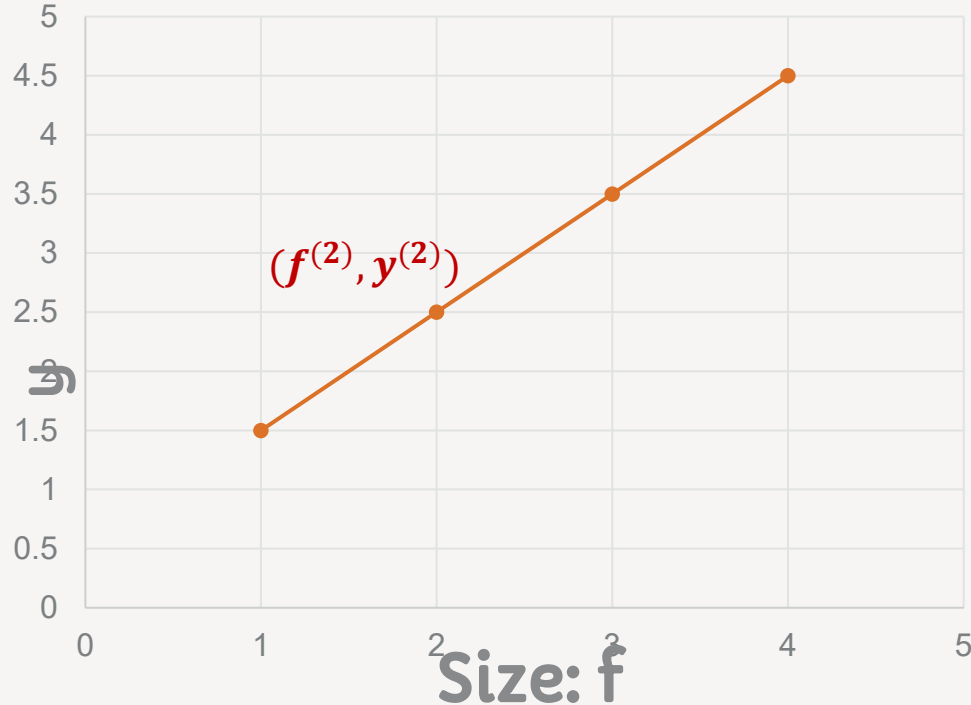


$$y = af$$

How to convert it to matrix-vector multiplication?

$$Ax = b$$

Linear Equation with offset



$$y = af + c$$

How to convert it to matrix-vector multiplication?

$$Ax = b$$

House Features

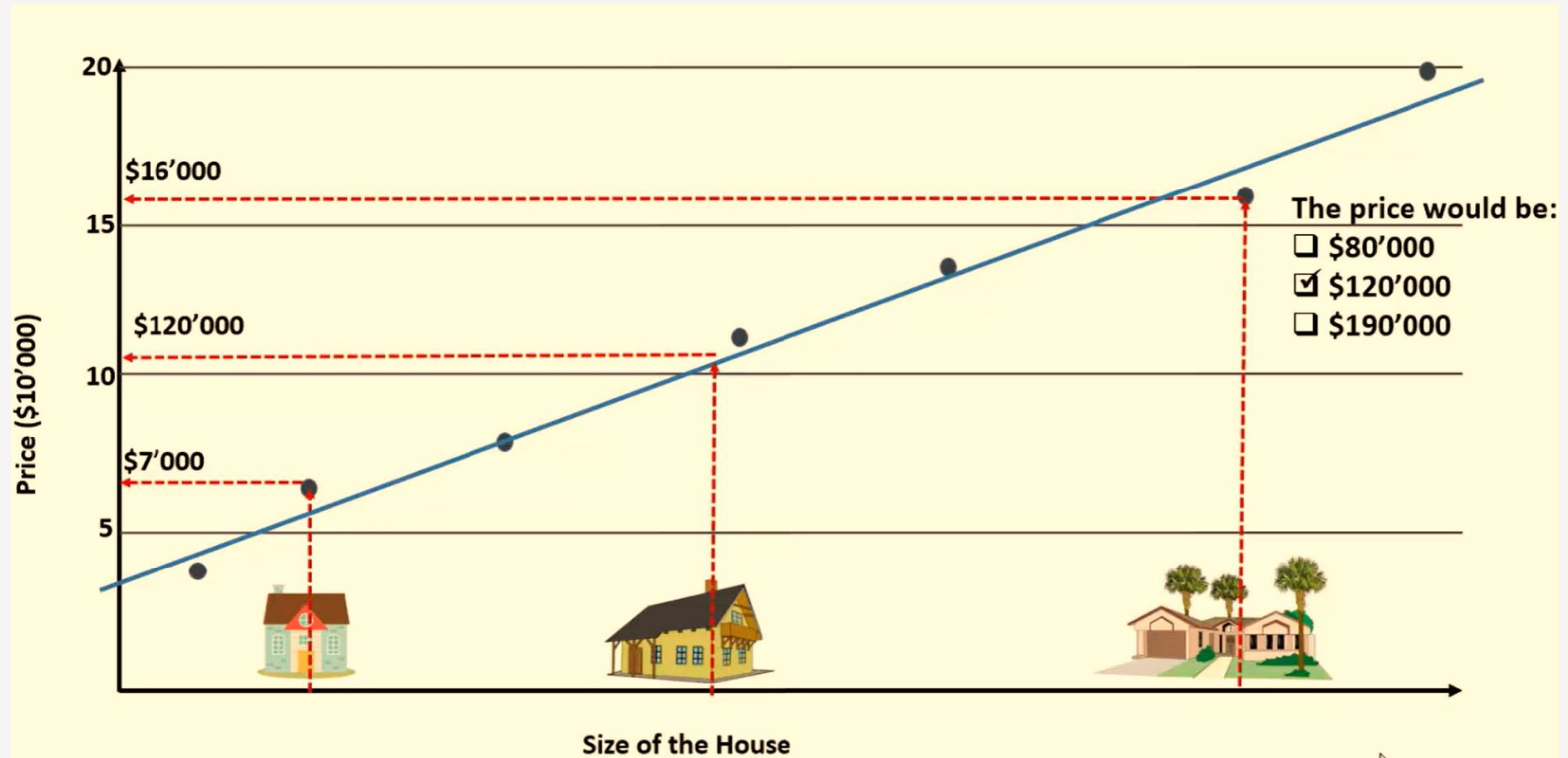
- #Room
- Size
- #Bedroom
- Age
- Address features: Street, Alley, ...
- Size of part1, part2, part3, part4
- Floors
- #Bathrooms

N number of training data with M features:

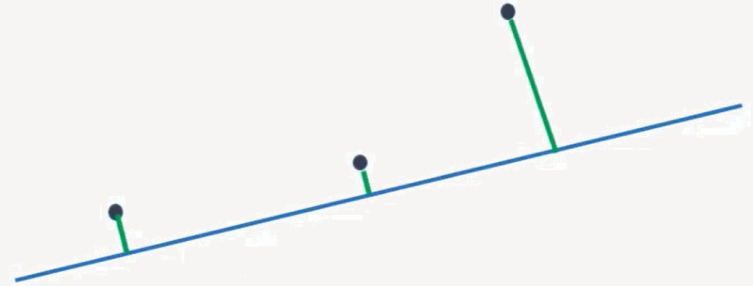
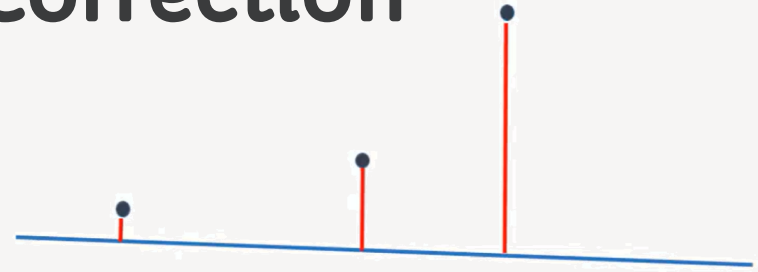
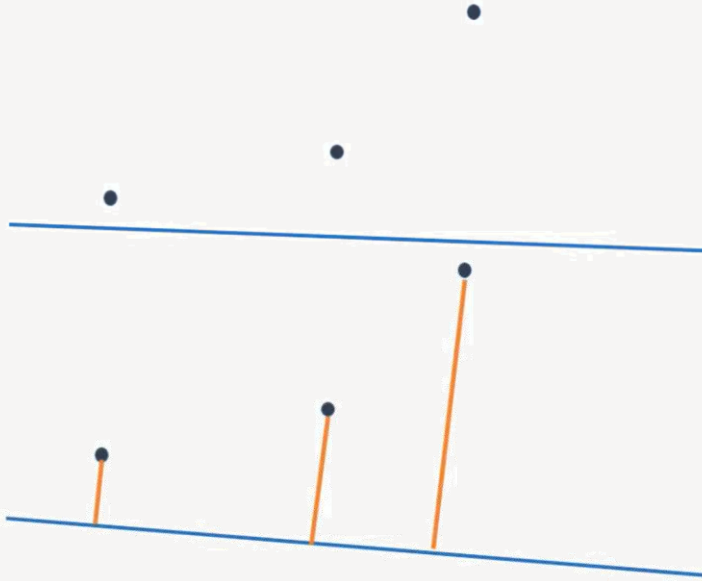
$$A_{N \times M} x_{M \times 1} = b_{N \times 1}$$

Linear Equation

Least Squares Error Correction



Least Squares Error Correction



Error 1: 

Error 2: 

Error 3: 

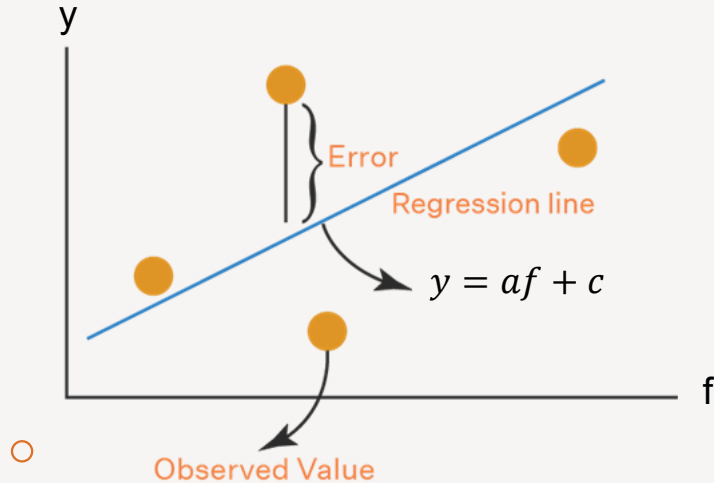
Least Squares

Least Square Method

- Objective for “m” features:

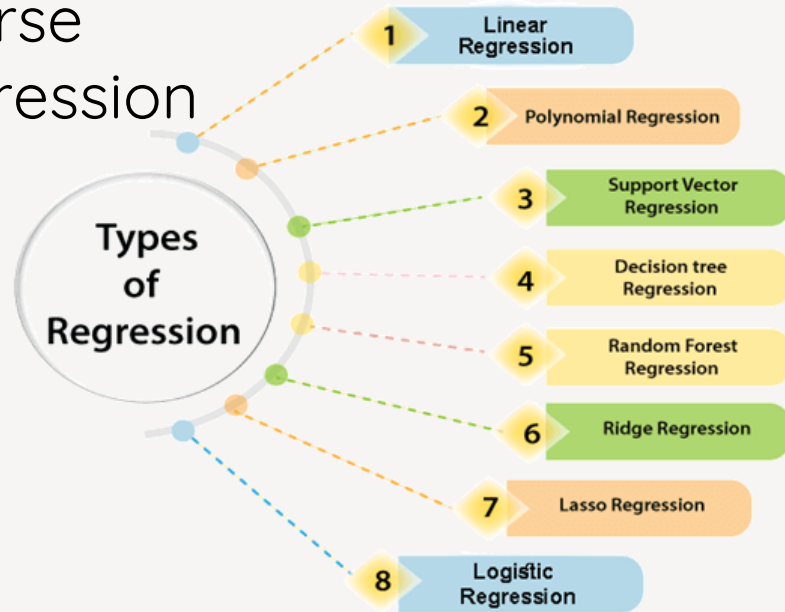
$$\hat{y} = a_1 f_1 + a_2 f_2 + \cdots + a_m f_m + c$$

$$\min ||y - \hat{y}||$$



Linear Algebra and Machine Learning Application

- $Ax = b \rightarrow x = A^{-1}b$ Inverse of Matrix/Pseudo Inverse
- Regression





02

Road Map

Step by Step



